



SCOPE

DA3 - Project Review III

Rubrics:

Description	Marks	Weightage
Front-End design with DB connectivity (Implementation)	10	10
Viva	5	
Report	5	

Objective:

- To design and implement a system that integrates a frontend interface with a backend database using JDBC/ ODBC/ API-based DB connectivity, enabling students to perform meaningful and efficient database operations.

REPORT FORMAT:

Database Connectivity using ODBC/ JDBC/ API-based Techniques

1. Title Page

- Project Title
- Course Title: Database Systems
- Class Number: CH2025260501233 / CH2025260501238
- Semester: Winter 2025-26
- Slot: A1+TA1 / A2+TA2
- Student Registration number and name
- Course Faculty: **Dr. L.M. Jenila Livingston**

2. Abstract

Brief overview of the project:

- Purpose of the system
- Technologies used (Frontend + Backend + DB + JDBC/ODBC/API based)
- Key functionalities

3. Introduction

- Background of the problem
- Need for database connectivity
- Objective of the project

4. System Overview

- Description of the application
- Target users
- Scope of the system

5. Technologies Used

- **Frontend:** HTML, CSS, JavaScript / React
- **Backend:** Java (Servlet/JSP) / Python (Flask/Django) / Node.js
- **Database:** MySQL / PostgreSQL / Oracle
- **Connectivity:** JDBC / ODBC/ API based DB connectivity

6. Database Design

6.1 ER Diagram

- Entities and relationships

6.2 Table Structure

Include:

- Table Name
- Attributes
- Data Types
- Constraints (PK, FK, NOT NULL)

Example:

Student (sid INT PRIMARY KEY, name VARCHAR(50), dept VARCHAR(20))

7. Database Connectivity Implementation (JDBC / ODBC/ API based)

7.1 Steps Involved

- Load Driver
- Establish Connection
- Create Statement
- Execute Query
- Process Result
- Close Connection

7.2 Sample Code

```
Connection con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/dbname","root","password");
```

```
PreparedStatement ps = con.prepareStatement("SELECT * FROM student");
ResultSet rs = ps.executeQuery();
```

8. Functional Modules

Explain each module clearly:

- User Registration/Login
- Data Entry Module
- Update/Delete Module
- Search/View Module

9. Implementation and Results:

Frontend Interface

- Screenshots of UI
- Explanation of user interaction

Backend Logic with CRUD operations

- How requests are handled
- Validation techniques (input validation)
- Query execution flow
- **CRUD Operations**
 - **Create** – Insert data
 - **Read** – Retrieve data
 - **Update** – Modify records
 - **Delete** – Remove records

10. Challenges Faced

- Connectivity issues
- Query errors
- UI-backend integration problems

11. Conclusion: Summary of work

12. Future Enhancements

- Security improvements
- Cloud database integration
- Advanced UI etc.